An All-Index Minimum Formula for the Denominator of the Mean of the Divisors

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Abstract

We give an exact finite arithmetic minimum formula for every term of OEIS A346644. Every integer whose divisor mean has reduced denominator n has a divisor with that same denominator whose exponent-plus-one values are supported on the primes of n and whose local prime-power means are reduced. This normal form yields a triangular allocation formula, including all cross-factor cancellations, and an exact evaluator that searches factorizations rather than consecutive integers. An elementary construction supplies an explicit finite bound without Dirichlet's theorem. The earlier prime-power partition formulas become special cases; in particular $a(2^s) = 2^{2^s-1}$. The all-index expression retains finite minimization and is not claimed to be an elementary closed form without it.

1 Definition

Let $\sigma(K)$ be the sum of the positive divisors of K and $\tau(K)$ their number. Define

$$a(n) = \min\{K \geq 1 : \operatorname{den}(\sigma(K)/\tau(K)) = n\}.$$

OEIS A346644 [1] records this sequence with zero as an escape value and conjectures that every index has a witness. The existence result below is elementary; the subsequent normal form and allocation formula determine the minimum itself for every index.

2 An elementary witness for every index

Write v_q for the q-adic valuation. For $m \geq 1$ put $S_p(m) = 1 + p + \cdots + p^{m-1}$, so the mean of the divisors of p^{m-1} is $S_p(m)/m$.

Lemma 1 (Upward numerator primes)**.** *Let q, p be primes, $p \not\equiv 1 \pmod q$, and $b \geq 1$. Every prime divisor of $S_p(q^b)$ is strictly greater than q.*

Proof. The sum is 1 modulo p. It is also 1 modulo q: if $p = q$ this is immediate, and otherwise Fermat's theorem gives $p^{q^b} \equiv p \pmod q$, with $p - 1$ invertible modulo q. If a prime r divides both the sum and $p - 1$, reducing the sum modulo r gives $r \mid q^b$, contradicting the preceding observation. For any other prime divisor r, the order of p modulo r is a nontrivial divisor of q^b. Hence $q \mid r - 1$, and $r > q$. □

Theorem 1 (Explicit elementary upper bound)**.** *Write $n = \prod_{i=1}^{s} q_i^{a_i}$ with $q_1 < \cdots < q_s$, and let ℓ_i be the i-th prime. Define $U_0 = 1$ and, successively,*

$$b_i = a_i + v_{q_i}(\sigma(U_{i-1})), \qquad U_i = U_{i-1}\ell_i^{q_i^{b_i}-1}. \tag{1}$$

Then $\operatorname{den}(\sigma(U_s)/\tau(U_s)) = n$, so $a(n) \leq U_s$. For $n = 1$ use $U_0 = 1$.

Proof. Since $\ell_i \le q_i$, the new base prime is not 1 modulo q_i and has not been used before. Its divisor count is $q_i^{b_i}$, and by Lemma 1 its divisor sum has no prime factor at most q_i. It therefore leaves all smaller denominator valuations unchanged. The previous divisor count is supported on $q_1, \dots, q_{i-1}$, so the new net q_i-valuation of the denominator is exactly $b_i - v_{q_i}(\sigma(U_{i-1})) = a_i$. Later factors do not change this valuation. The final divisor count has no primes outside those of n, which proves the assertion. This construction needs no theorem about primes in arithmetic progressions. □

3 A divisor-preserving normal form

For primes q, p and $m \ge 1$, define the signed local contribution

$$c_q(p, m) = v_q(m) - v_q(S_p(m)).$$

For $K = \prod p_i^{m_i - 1}$, the valuation of q in its reduced denominator is $\max(0, \sum_i c_q(p_i, m_i))$.

Lemma 2 (Triangular dependence)**.** *For an odd prime q, the contributions are as follows, with $h = \mathrm{ord}_q(p)$ in the last two cases:*

$$c_q(p, m) = \begin{cases} v_q(m), & p = q, \\ 0, & p \equiv 1 \pmod q, \\ v_q(m), & p \not\equiv 0, 1 \pmod q,\ h \nmid m, \\ -v_q(p^h - 1), & p \not\equiv 0, 1 \pmod q,\ h \mid m, \end{cases} \tag{2}$$

For $q = 2$, $c_2(2, m) = v_2(m)$, while for odd p the contribution is 0 when m is odd and $1 - v_2(p+1)$ when m is even. Deleting factors of primes smaller than q from m cannot decrease $c_q(p, m)$; deleting factors of primes greater than q cannot change it.

Proof. The cases $p = q$ and $h \nmid m$ have numerator valuation zero. For $p \equiv 1 \pmod q$, the odd-prime lifting identity gives $v_q(S_p(m)) = v_q(m)$. If $h \mid m$ and $p \not\equiv 0, 1 \pmod q$, lifting gives $v_q(S_p(m)) = v_q(p^h - 1) + v_q(m)$. The stated 2-adic identity follows from the usual even-exponent lifting formula, and an odd number of odd summands is odd. Finally $h \mid q - 1$. Deleting a smaller prime can only change $h \mid m$ to $h \nmid m$, replacing a negative contribution by a nonnegative one. Deleting a larger prime changes neither this condition nor $v_q(m)$. The assertions for $q = 2$ follow from parity. □

Theorem 2 (Reduced-factor normal form)**.** *If $\mathrm{den}(\sigma(K)/\tau(K)) = n$, there is a divisor $K' = \prod p_i^{m_i - 1}$ of K with the same reduced denominator such that*

1. *every prime divisor of every m_i divides n;*
2. *$\gcd(m_i, S_{p_i}(m_i)) = 1$ for every i.*

Consequently the least value $a(n)$ itself has both properties.

Proof. Allow $m_i = 1$ to retain a base with exponent zero. Process the prime divisors of the original $\tau(K) = \prod m_i$ in increasing order, only replacing m_i by divisors of themselves.

At $q = 2$, remove all 2-parts from the m_i attached to odd bases. If $2 \mid n$, retain exactly $v_2(n)$ powers of 2 in the m_i for base 2; the original exponent has enough capacity, since every odd base contributes nonpositively at 2. If $2 \nmid n$, remove its 2-part as well. All local means are now reduced at 2, and the denominator has the required 2-valuation. Larger denominator primes have not lost any signed contribution, by Lemma 2.

For an odd prime q, classify the current factors using (2). Call the factors in the first or third case active. The fourth-case contributions are constants $-\alpha_i$, and the second-case contributions are zero. Remove the q-parts of m_i from every inactive factor; this changes none of those signed contributions. If $q \nmid n$, also remove all active q-parts, leaving a nonpositive sum and hence denominator valuation zero. If $q \mid n$, put $a = v_q(n)$. Earlier steps have only increased the signed q-sum, so the active capacities satisfy

$$\sum_{i \text{ active}} v_q(m_i) \ \ge \ a + \sum_{i \text{ fourth}} \alpha_i.$$

Retain any allocation of exactly the right-hand side among these capacities. The resulting signed sum is exactly a. Active numerator sums are coprime to q, and the other factors now have q-free m_i, so every local mean is reduced at q. Later steps leave this conclusion and the total q-contribution unchanged. Earlier denominator valuations also remain unchanged.

At the end, the divisor count is supported on the primes of n, all local means are reduced, and the total reduced denominator is n. All exponent changes decreased m_i, so $K' \mid K$. Applying this to a least K forces $K' = K$. □

Corollary 1. *The odd part of $a(n)$ is always a square. If n is odd, $a(n)$ is itself a square. If $2^a \| n$ with $a \geq 1$, then $2 \mid a(n)$ and*

$$v_2\big(v_2(a(n)) + 1\big) = a.$$

Proof. An odd base with even m_i violates local coprimality at 2. Thus every odd-base exponent $m_i - 1$ is even. If n is odd, all m_i are odd by the support restriction. In the remaining case only base 2 can supply the required 2-valuation, and all other local contributions at 2 are zero. □

4 The exact minimum for every index

The following is a finite arithmetic formula using multiplicative orders, valuations, weak compositions, and a finite minimum. Its stages depend only on the factorization of n, not on previous sequence values. It is not asserted to be an elementary expression without finite minimization.

For $n > 1$, fix the explicit witness $U = U_s$ of (1) and set $L = \lfloor \log_2 U \rfloor$. Take each tuple of distinct primes

$$2 \leq p_1 < \cdots < p_t \leq U, \qquad 1 \leq t \leq L. \tag{3}$$

Start with $m_1 = \cdots = m_t = 1$. For $q = q_1, \ldots, q_s$ in increasing order, write $a = v_q(n)$ and use the current m_i to form

$$\begin{aligned} G_q &= \{i : p_i = q\} \cup \{i : p_i \not\equiv 0, 1 \pmod q,\ \operatorname{ord}_q(p_i) \nmid m_i\}, \\ N_q &= \{i : p_i \not\equiv 0, 1 \pmod q,\ \operatorname{ord}_q(p_i) \mid m_i\}, \\ B_q &= a + \sum_{i \in N_q} v_q(p_i^{\operatorname{ord}_q(p_i)} - 1). \end{aligned}$$

For $q = 2$, these definitions simply give $G_2 = \{i : p_i = 2\}$, $N_2 = \varnothing$, and $B_2 = v_2(n)$. For every weak composition

$$b_i \geq 0 \ (i \in G_q), \qquad \sum_{i \in G_q} b_i = B_q, \tag{4}$$

make a branch with $m_i \leftarrow m_i q^{b_i}$ on G_q, leaving the other m_i unchanged. If G_q is empty, there is no branch. Let $\mathcal{C}(n)$ consist of all products $\prod p_i^{m_i - 1}$ obtained after the last stage, over the tuples (3) and all the specified weak compositions. Exponents zero may be discarded.

Theorem 3 (All-index minimum formula)**.** *The set $\mathcal{C}(n)$ is finite and nonempty, every member has reduced divisor-mean denominator n, and*

$$\boxed{a(1) = 1, \qquad a(n) = \min \mathcal{C}(n) \quad (n > 1).} \tag{5}$$

Any verified witness may replace U in (3).

Proof. Before the q-stage, m_i has only prime factors smaller than q. Adding q-powers changes the active contributions to b_i; the inactive contributions remain zero or the constants occurring in B_q. Equation (4) therefore makes the signed sum exactly a. At 2 only base 2 receives an even m_i, so the same conclusion holds. All local means remain reduced at the processed prime. Later stages have larger primes and cannot change any earlier signed contribution. Since the final divisor count is supported on the primes of n, every output has denominator exactly n.

There are finitely many prime tuples, finitely many stages, and finitely many weak compositions at each stage. Thus the set is finite. By Theorem 1, a least value $K = a(n) \leq U$ exists. Its number of distinct prime factors is at most $\lfloor \log_2 U \rfloor$, and each base prime is at most U. By Theorem 2, its exponent-plus-one values have the required support and local coprimality. At every stage, local coprimality forces $b_i = 0$ on inactive factors, while its denominator valuation forces exactly (4). Its factorization is consequently one of the branches in $\mathcal{C}(n)$. This proves nonemptiness, completeness for the least value, and the formula. Only $a(n) \leq U$ was needed, giving the last assertion. □

4.1 Exact evaluation without scanning successive integers

The implementation searches increasing tuples of base primes with m_i supported on the primes of n and checks local coprimality. For a partial product V, let c_q be its current signed contribution and put

$$E = \sum_{q \mid n} \max(0, v_q(n) - c_q)(q - 1).$$

Any completion needs at least E additional prime factors counted with multiplicity: for every positive m supported on these primes,

$$m - 1 \geq \sum_{q \mid n} v_q(m)(q-1),$$

by expanding the product of its prime factors, and additional numerator valuations can only increase the required divisor-count contributions. If the partial denominator is not already n, at least one additional factor is required, even if $E = 0$. When the next base prime is p, the completion is therefore at least $Vp^{\max(1,E)}$. This supplies a rigorous stopping bound for each prime loop. Exponents obey $m_i \leq 1 + \lfloor \log_{p_i}((U-1)/V) \rfloor$ when seeking an improvement to a known witness U. Every cutoff uses exact integer comparisons.

For smaller starting bounds, the implementation may split a q-budget among several unused primes not congruent to 1 modulo q, using Lemma 1. Such primes are elementary to obtain for odd q: if a finite list exhausted them, a prime divisor not congruent to 1 modulo q of q times their product minus 1 would be missing from the list. The case $q = 2$ uses only base 2. These choices affect speed, not the completeness theorem; every starting bound is independently checked.

For example, $n = 30$ and bases $(2, 3, 5)$ admit the successive budgets $B_2 = 1$, $B_3 = 2$, and $B_5 = 1$. Choosing allocations $(1, 0, 0)$, $(0, 1, 1)$, and $(1, 0, 0)$ in base order, with zeros on inactive bases, gives final $m = (10, 3, 3)$, hence

$$a(30) = 2^9 3^2 5^2 = 115200.$$

The exact evaluator excludes all smaller candidates using the proved normal form and bounds. In particular, the minimum cannot be obtained by requiring $\tau(K) = n$: here $\tau(K) = 90$ and cross-factor cancellation is essential. Further examples are $a(40) = 6480000$, $a(42) = 1843200$, $a(45) = 3600$, and $a(60) = 4706440200$.

5 Exact least values at every prime power

A factor p^e of K contributes

$$c_q(p, e+1) = v_q(e+1) - v_q(1 + p + \cdots + p^e)$$

to the denominator valuation before taking its positive part.

Lemma 3. *If $c_q(p, e+1) > 0$, then $p \not\equiv 1 \pmod q$ and $c_q(p, e+1) \leq v_q(e+1)$. For any prime $p \not\equiv 1 \pmod q$ and integer $b \geq 1$,*

$$\operatorname{den}\left(\frac{1 + p + \cdots + p^{q^b - 1}}{q^b}\right) = q^b. \tag{6}$$

Proof. The inequality follows immediately from the nonnegative numerator valuation. If q is odd and $p \equiv 1 \pmod q$, lifting the exponent gives $v_q((p^{e+1} - 1)/(p-1)) = v_q(e+1)$, so $c_q(p, e+1) = 0$. For $q = 2$ and odd p, if e is even both valuations are zero; if e is odd, the elementary 2-adic lifting formula gives

$$v_2((p^{e+1} - 1)/(p-1)) = v_2(p+1) + v_2(e+1) - 1 \geq v_2(e+1).$$

Thus positive contribution again requires $p \not\equiv 1 \pmod q$. Finally $p^{q^b} \equiv p \pmod q$ by Fermat's theorem, and $p - 1$ is invertible modulo q. The numerator sum in (6) is therefore 1 modulo q, proving the asserted reduced denominator. □

Theorem 4 (Partition formula)**.** *Fix an odd prime q, and list the primes not congruent to 1 modulo q as $\ell_1(q) < \ell_2(q) < \cdots$. For every $s \geq 1$,*

$$a(q^s) = \min_{\substack{b_1 \geq \cdots \geq b_t \geq 1 \\ b_1 + \cdots + b_t = s}} \prod_{i=1}^{t} \ell_i(q)^{q^{b_i} - 1}. \tag{7}$$

The minimum is over the ordinary integer partitions of s. For $q = 2$,

$$a(2^s) = 2^{2^s - 1} \qquad (s \geq 0). \tag{8}$$

Consequently $a(q) = 2^{q-1}$ for every prime q.

Proof. Every candidate in (7) has divisor count q^s and divisor sum coprime to q, by (6). Its reduced denominator is exactly q^s.

Conversely, let $K = \prod p_i^{e_i}$ have reduced denominator q^s. The sum of all $c_q(p_i, e_i + 1)$ is s, so the sum over its positive terms is at least s. By Lemma 3, the corresponding primes are admissible and their values $a_i = v_q(e_i + 1)$

have sum at least s. Choose integers $0 \le b_i \le a_i$ with total s, discard the zeros, and replace each exponent by $q^{b_i} - 1 \le e_i$. The resulting integer $K' \le K$ has denominator exactly q^s by (6). Thus it suffices to minimize such products.

For a fixed list of positive b_i, use the smallest admissible primes and assign larger exponents to smaller primes. The latter follows from $x^A y^B \le x^B y^A$ when $x \le y$ and $A \ge B$. This gives (7). There are sufficiently many admissible primes by the elementary argument at the end of Section 4.

For $q = 2$, the only admissible prime is 2. The same replacement argument requires the exponent of 2 to be at least $2^s - 1$. The integer 2^{2^s-1} attains denominator 2^s, since its divisor sum is odd; this proves (8), including $a(1) = 1$. For odd prime q, the only partition of 1 is (1) and $\ell_1(q) = 2$. □

For example $a(9) = \min(2^8, 2^2 3^2) = 36$ and $a(25) = \min(2^{24}, 2^4 3^4) = 1296$. At denominator 81 the partition $(2, 1, 1)$ gives $2^8 3^2 5^2 = 57600$, smaller than the value 108900 from $(1, 1, 1, 1)$; comparison with the other three partitions proves $a(81) = 57600$.

6 Verification and scope

The standard-library program `minimum.py` evaluates the exact minimum using the normal form and integer branch bounds. Its separate `triangular_candidates` generator implements the allocation formula directly for fixed prime tuples. The independent audit in `verify_minimum.py` checks divisor-preserving normalization for all $K \le 50000$, and compares every minimum occurring at $K \le 10^6$ against a sieve built by summing actual divisors. All 39 published terms agree, and complete bounded candidate sets are compared in both directions with direct enumeration. The older `verify.py` remains an independent control for the original witnesses and prime-power cases.

The universal claims follow from the proofs, not from these finite checks. The formula uses finite minimization, valuations and weak compositions; no simpler elementary expression is asserted. No OEIS editorial acceptance or exhaustive literature-priority review is claimed.

References

[1] B. Cloitre, OEIS A346644, https://oeis.org/A346644, accessed October 26, 2026.